LESSON 6: ANTIHYPERTENSIVE DRUGS

Blood pressure, hypertension, and the 4 drug groups by where they act

PHARMACOLOGY | FINALS: ANTIHYPERTENSIVE, ANTI-INFLAMMATORY AND GI DRUGS

How do I keep the antihypertensives straight?

Start from **BP = CO × SVR**. Every drug lowers **cardiac output**, **vascular resistance** or **blood volume** at one of **4 sites**: **arterioles**, **venules**, **heart**, **kidneys**. The slides group them into **4 classes**: **diuretics** (kidney, volume), **sympatholytics** (block the sympathetic push), **direct vasodilators** (open the vessels) and **angiotensin blockers** (stop the RAAS). The **name endings** give away most classes.

What Dr. Orig stressed in class

Memorize the BP categories: he will give BP readings to interpret (120/80 is already **stage 1**; the higher category wins). **Memorize the four loop diuretics**: **furosemide (Lasix)** is the most used, **bumetanide** the most potent, **ethacrynic acid** the least. Loops are for **acute pulmonary edema**, no longer for hypertension; **thiazides** are the usual diuretic for hypertension. **Methyldopa and clonidine are α2 AGONISTS**; methyldopa is only for pregnancy. **Amlodipine** is the most common CCB. His usual question: **ACE inhibitor cough → switch to an ARB**; the usual combination is **losartan + amlodipine**. He skipped the hormone table, the 4-sites figure and the thiazide dose table.

CIRCULATION REVIEW

- **Blood flow**: body → **right heart** → **lungs** (pulmonary circuit) → **left heart** → body (systemic circuit)
- **Arteries** carry blood **away from** the heart; **veins** carry it **back** and have **valves**; **capillaries** = one layer of **endothelial cells** (gas exchange)
- Vessel wall layers: **tunica interna** (endothelium, subendothelial layer, internal elastic lamina) → **tunica media** (smooth muscle, external elastic lamina) → **tunica externa**

BLOOD PRESSURE

- **Blood pressure** – the pressure exerted by blood on the **walls of a blood vessel**
- **BP = CO × SVR** (cardiac output × systemic vascular resistance)
- **CO = HR × SV** (heart rate × stroke volume)
- **Systolic (SBP)** – the **upper** number (150 in 150/90); pressure while the heart is **contracting** (at the heartbeat)
- **Diastolic (DBP)** – the **lower** number (90); pressure while the heart is **relaxed** (between beats)

WHAT RAISES MEAN ARTERIAL PRESSURE (MAP)

- ↑ **Cardiac output**, from:
  - ↑ **Heart rate** – less parasympathetic, more **sympathetic** impulses and **adrenal medulla** hormones
  - ↑ **Stroke volume** – more **venous return** (more blood volume, skeletal muscle pump, respiratory pump, venoconstriction)
- ↑ **Systemic vascular resistance**, from:
  - ↑ **Blood viscosity** (more RBCs, as in **polycythemia**)
  - ↑ **Total vessel length** (bigger body size, as in **obesity**)
  - ↓ **Vessel radius** (**vasoconstriction**)

HORMONES THAT CONTROL BP

> **Note:** Dr. Orig **skipped** this table and the 4-sites figure below in class and said to **focus on the drugs**. They are kept here for reference.

- **Raise heart rate and contractility** (↑ BP): **norepinephrine, epinephrine**
- **Vasoconstrict** (↑ BP): **angiotensin II, ADH**, norepinephrine (α1), epinephrine
- **Vasodilate** (↓ BP): **ANP**, epinephrine (β2, in heart and skeletal muscle arterioles), **nitric oxide**
- **Raise blood volume** (↑ BP): **aldosterone, ADH**
- **Lower blood volume** (↓ BP): **ANP** (atrial natriuretic peptide)

Note: Memory tip: ANP is the only hormone on the list that **lowers** BP in two ways (vasodilation and less volume). **ADH** and **angiotensin II** raise it in two ways.

4 ANATOMIC SITES OF BP CONTROL

- **1. Arterioles** – **resistance** vessels
- **2. Venules** – **capacitance** vessels
- **3. Heart** – **pump output**
- **4. Kidneys** – **volume** (through **renin** → **angiotensin** → **aldosterone**)
- All four are driven by the **CNS sympathetic nerves**

HYPERTENSION

- Diagnosed on **repeated, reproducible** measurements of elevated BP, not one reading
- **Primary (essential)** – **no specific cause** (most patients)
- **Secondary** – has a **specific etiology**

BP CATEGORIES (SYSTOLIC / DIASTOLIC, MMHG)

- **Normal** – below **120 and** below **80**
- **Elevated** – **120–129 and** below 80
- **Stage 1** – **130–139 or 80–89**
- **Stage 2** – **≥140 or ≥90**
- **Hypertensive crisis** – above **180 and/or** above **120**

Note: Only "Normal" and "Elevated" need **both** numbers in range (**and**); stages 1 and 2 need **either** number (**or**). So 125/85 is **Stage 1** because of the diastolic.

THE 4 CLASSES OF ANTIHYPERTENSIVES

- **1. Diuretics**
- **2. Sympathoplegic (sympatholytic) agents**
- **3. Direct vasodilators**
- **4. Agents that block the production or action of angiotensin**

1. DIURETICS

- Lower BP by **depleting body sodium stores**
- Lower BP by **10–15 mmHg** in most patients
- Used **alone** in **mild or moderate essential hypertension**

SITES OF ACTION (ALONG THE NEPHRON)

- **1. Proximal convoluted tubule** – **acetazolamide** (carbonic anhydrase inhibitor)
- **2. PCT and descending limb** – **osmotic diuretics** (**mannitol**)
- **3. Thick ascending limb (TAL) of the loop of Henle** – **loop diuretics** (furosemide, torsemide)
- **4. Distal convoluted tubule** – **thiazide-type** (HCTZ, chlorthalidone, metolazone)
- **5. Collecting duct** – **potassium-sparing** (spironolactone, triamterene)

Note: Memory tip: go down the nephron in **A-O-L-T-K** order: "**A**ll **O**ur **L**oops **T**ake **K**+": **A**cetazolamide, **O**smotic, **L**oop, **T**hiazide, **K**-sparing.

LOOP DIURETICS (THE STRONGEST)

- Inhibit **NaCl reabsorption in the TAL**
- **MOA:** inhibit **NKCC2**, the luminal **Na+/K+/2Cl- cotransporter** in the TAL
- **2 prototypes: furosemide and ethacrynic acid**; also **bumetanide, torsemide**
- **Uses: acute pulmonary edema** and other edematous conditions; **hyperkalemia**; **acute renal failure**
- **Toxicity: ototoxicity**; **allergic reactions** (sulfa)
- Daily oral doses: **bumetanide 0.5–2 mg**, **torsemide 5–20 mg**, **furosemide 20–80 mg**, **ethacrynic acid 50–200 mg**

Note: Memory tip: the **ear has a loop too** (the cochlea), so **loop** diuretics hurt the **ear** (ototoxicity). Loops **throw out K+**, which is why they treat **hyperkalemia**.

Note: Textbook extra: **ethacrynic acid** is the loop diuretic **without a sulfa group**, the one for patients allergic to sulfa drugs.

THIAZIDE DIURETICS

- **MOA:** block the **Na+/Cl- cotransporter (NCC)** in the **DCT**
- **Prototype: hydrochlorothiazide (HCTZ)**, 25–100 mg once a day
- **Uses: hypertension**, **heart failure**
- **Toxicity: hyperlipidemia**, **hyponatremia**, **gout**, **allergic reactions**

- **Thiazide-like** (not thiazides, but **sulfonamides** that act the same): **chlorthalidone, indapamide, metolazone, quinethazone**
- Taken in **two divided doses**: **chlorothiazide** (0.5–2 g) and **hydroflumethiazide**; all others **once a day**

Note: Memory tip: thiazide toxicity = **LUNA**: **L**ipids up, **U**ric acid up (gout), **Na+** down, **A**llergy (sulfa).

Note: Loop vs thiazide: loop = **NKCC2** in the **TAL** (strongest, for **edema**); thiazide = **NCC** in the **DCT** (for **hypertension**).

POTASSIUM-SPARING DIURETICS

- Prevent **K+ secretion** by antagonizing **aldosterone** in the **collecting tubules**
- **1. Aldosterone antagonists** – **spironolactone, eplerenone** (block the mineralocorticoid receptor)
- **2. Epithelial Na+ channel (ENaC) inhibitors** – **amiloride, triamterene**
- **Uses: hyperaldosteronism**, **heart failure**
- **Toxicity: hyperkalemia**; **gynecomastia (spironolactone)**

Note: Memory tip: they **spare K+**, so K+ builds up: **hyperkalemia**. **Spiro**nolactone is a steroid look-alike, so it causes **gynecomastia**; **eplerenone** is the more selective one.

2. SYMPATHOLYTIC (SYMPATHOPLEGIC) DRUGS

- Sympathetic discharge → **α1** on **blood vessels** (vasoconstriction) + **β1** on the **heart** (↑ rate and force, ↑ afterload) → ↑ BP. These drugs cut that push.
- **1. Methyldopa**
- **2. Clonidine**
- **3. Beta-blockers**
- **4. Prazosin** and other **alpha-1 blockers**
- **5. Phentolamine, phenoxybenzamine**

Note: From Lesson 4: **methyldopa** and **clonidine** are **central α2 agonists** (the brain lowers sympathetic outflow). **Methyldopa** = **M**om's drug (**gestational hypertension, preeclampsia**). **-olol** = β-blockers; **-zosin** = α1 blockers; **phe**ntolamine and **phe**noxybenzamine are for **phe**ochromocytoma.

Note: Textbook extra: stopping **clonidine** suddenly causes **rebound hypertension**.

3. DIRECT VASODILATORS

- **Oral: hydralazine**, **minoxidil**
- **Parenteral (IV): nitroprusside**, **fenoldopam**

MECHANISMS (KATZUNG TABLE 11-3)

- **Release nitric oxide** (from the drug or the endothelium) – **nitroprusside, hydralazine**, nitrates, histamine, acetylcholine
- **Reduce calcium influx** – **verapamil, diltiazem, nifedipine** (the CCBs)
- **Hyperpolarize the membrane by opening K+ channels** – **minoxidil, diazoxide**
- **Activate dopamine receptors** – **fenoldopam**

Note: Memory tip: Minoxidil and **Dia**zoxide **open the K+ door**. **Fenol-DOPA-m** works on **DOPA**mine receptors. The slide shows **topical minoxidil 5%** (Regroe): its side effect of **hair growth** became a **hair-regrowth** product.

CALCIUM CHANNEL BLOCKERS (CCBS)

- **MOA:** inhibit **Ca2+ influx into arterial smooth muscle** cells → **dilated arteries**, lower BP
- **Non-dihydropyridines: verapamil, diltiazem** – also **slow the SA node** (heart rate), so the ventricle fills fully and the heart's workload drops
- **Dihydropyridines (-dipine): amlodipine, felodipine, nifedipine, nicardipine** – mainly the **vessels**

Note: Memory tip: "**-dipine dilates**" (vessels). **V**erapamil and **D**iltiazem also **V**isit the heart and **D**ecrease its rate.

4. INHIBITORS OF ANGIOTENSIN (RAAS)

THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

- Trigger: **drop in BP** or **drop in fluid volume**
- **Kidney** releases **renin** → renin turns **angiotensinogen** (from the **liver**) into **angiotensin I**
- **ACE** (angiotensin-converting enzyme, from the **lungs**) turns angiotensin I into **angiotensin II**
- **Angiotensin II** → **vasoconstriction** (narrows vessels directly) + **adrenal gland releases aldosterone**

- **Aldosterone** → kidneys **reabsorb salt (NaCl) and water** → more volume, higher BP

WHERE EACH DRUG HITS THE RAAS

- **Aliskiren** – blocks **renin**
- **ACE inhibitors** – block **ACE** (Ang I → Ang II)
- **ARBs** – block the **angiotensin II receptor** (both vasoconstriction and aldosterone release)
- **Spironolactone, eplerenone** – block **aldosterone's** effect
- **Bradykinin** side branch: **ACE is the same enzyme as kininase II**, which breaks down **bradykinin** (a **vasodilator** that also raises **prostaglandins**)

Note: Memory tip: go down the cascade **R-A-A-A**: **R**enin (ali**skiren**) → **A**CE (-**pril**) → **A**ngiotensin II receptor (-**sartan**) → **A**ldosterone (spironolactone).

ACE INHIBITORS (-PRIL)

- **MOA:** inhibit the enzyme that **converts angiotensin I to angiotensin II** and **inactivates bradykinin**, so **Ang II falls** and **bradykinin rises**
- **Drugs: captopril, enalapril, ramipril**
- Used in **chronic kidney disease**: they **diminish proteinuria** and **stabilize renal function** (especially in **diabetics**)
- **Toxicity: dry cough**, **angioedema** (swollen lips, shown on the slide), **contraindicated in pregnancy**

Note: Memory tip: the extra **bradykinin** causes the **cough** and the **angioedema**. ACEi side effects = **ACE**: **A**ngioedema, **C**ough, **E**mbryo (no pregnancy).

ANGIOTENSIN RECEPTOR BLOCKERS (ARBS, -SARTAN)

- **MOA:** block the angiotensin II **type 1 (AT1)** receptor
- **Drugs: losartan, valsartan, candesartan, irbesartan, telmisartan**
- **More selective** than ACE inhibitors: **no effect on bradykinin**
- **AT1** effects (blocked): **vasoconstriction, sympathetic activation, cell proliferation, aldosterone release, renal sodium reabsorption** → atherosclerosis, hypertension
- **AT2** effects (left alone): **vasodilation, inhibition of cell growth, apoptosis**

Note: ACEi vs ARB: same result on angiotensin, but the ARB leaves **bradykinin** alone, so **no cough**. A patient with an ACEi cough is switched to an **ARB**. Both are **contraindicated in pregnancy**.

TREATMENT ALGORITHM (ADULTS)

- **BP ≥ 140/90** → advise **therapeutic lifestyle changes**: **low-salt diet, weight loss, exercise, smoking cessation, alcohol abstinence**
- **High risk?** (**HMOD**, CV disease, or CV-risk equivalent)
  - **Yes** → start drugs now
  - **No** → lifestyle for **3 months**; target reached → keep the lifestyle change; not reached → start drugs
- **First-line drugs** (alone or combined): **ACE inhibitor, ARB, CCB, thiazide diuretic**
- Targets not met after **3 months**:
  - On **monotherapy** → add a **2nd** first-line agent
  - Already on **2 drugs** → add **another** first-line drug
  - Still not at target on the combination → **refer**
- **Preferred combinations: ACEi/ARB + CCB** and **ACEi/ARB + HCTZ**
- **Beta-blockers** as **initial** therapy only with **coronary artery disease, acute coronary syndrome, high sympathetic drive**, or **pregnant women**
- **Resistant hypertension** – still uncontrolled on **3 drugs, one of them a diuretic** → refer to a **specialist** for work-up or **second-line** agents

Note: Memory tip: first line is **A-C-D**: **A**CEi or **A**RB, **C**CB, **D**iuretic (thiazide). **B** (beta-blocker) waits for a reason: a sick heart, too much sympathetic drive, or pregnancy. Never combine an **ACEi with an ARB** (both are "A").

Name endings at a glance

-thiazide thiazide · **-semide / -etanide** loop · **-pril** ACE inhibitor · **-sartan** ARB · **-kiren** renin inhibitor · **-dipine** CCB (dihydropyridine) · **-olol** β-blocker · **-zosin** α1 blocker

SUMMARY TABLE: ANTIHYPERTENSIVE DRUGS

Class (ending)	Drugs	Mechanism (site)	Uses	Toxicity
Loop diuretics (-semide, -etanide)	**furosemide**, **ethacrynic acid** (prototypes), **bumetanide**, **torsemide**	block **NKCC2** (Na+/K+/2Cl-) in the **TAL**	**acute pulmonary edema**, edema, **hyperkalemia**, **acute renal failure**	**ototoxicity**, **allergy** (sulfa)
Thiazides (-thiazide)	**HCTZ** (prototype); thiazide-like: **chlorthalidone, indapamide, metolazone**	block **NCC** (Na+/Cl-) in the **DCT**	**hypertension**, **heart failure**	**hyperlipidemia, hyponatremia, gout, allergy** (LUNA)
K+-sparing: aldosterone antagonists	**spironolactone**, **eplerenone**	block aldosterone in the **collecting tubule**	**hyperaldosteronism**, **heart failure**	**hyperkalemia**; **gynecomastia** (spironolactone)
K+-sparing: ENaC inhibitors	**amiloride**, **triamterene**	block the epithelial Na+ channel (**ENaC**)	with other diuretics to spare K+	**hyperkalemia**
Central α2 agonists	**methyldopa**, **clonidine**	brain lowers sympathetic outflow	hypertension; **methyldopa in pregnancy**	clonidine: **rebound hypertension** if stopped
β-blockers (-olol)	propranolol, **atenolol**, **metoprolol**, labetalol…	block **β1** → ↓ heart rate and force	first line only with **CAD, ACS, high sympathetic drive, pregnancy**	bradycardia; bronchospasm (nonselective)
α1 blockers (-zosin)	**prazosin**, doxazosin	block α1 → vessels relax	hypertension, BPH	**orthostatic hypotension**
α nonselective	**phentolamine**, **phenoxybenzamine**	block α1 + α2	**pheochromocytoma**	reflex tachycardia
Oral vasodilators	**hydralazine** (NO), **minoxidil** (K+ channel opener)	relax arteriolar smooth muscle	hypertension	minoxidil: **hair growth** (now a topical hair-regrowth drug)
Parenteral vasodilators	**nitroprusside** (NO), **fenoldopam** (dopamine receptor)	relax arterioles, given IV	**hypertensive emergencies**	–
CCBs: dihydropyridines (-dipine)	**amlodipine**, **felodipine**, **nifedipine**, **nicardipine**	block **Ca2+ influx** into arterial smooth muscle	**first line**	–
CCBs: non-dihydropyridines	**verapamil**, **diltiazem**	block Ca2+ influx; also **slow the SA node**	**first line**	–
ACE inhibitors (-pril)	**captopril**, **enalapril**, **ramipril**	block **ACE** → ↓ Ang II, ↑ **bradykinin**	**first line**; **CKD** (↓ proteinuria), esp. **diabetics**	**dry cough**, **angioedema**, **not in pregnancy**
ARBs (-sartan)	**losartan**, **valsartan**, **candesartan**, **irbesartan**, **telmisartan**	block the **AT1** receptor; **no bradykinin** effect	**first line**; when an ACEi causes cough	not in pregnancy
Renin inhibitor (-kiren)	**aliskiren**	blocks **renin** (top of the RAAS)	hypertension	–

SUMMARY TABLE: BLOOD PRESSURE CATEGORIES			
Category	**Systolic**		**Diastolic**
Normal	below 120	**and**	below 80
Elevated	120–129	**and**	below 80
Hypertension stage 1	130–139	**or**	80–89
Hypertension stage 2	140 or higher	**or**	90 or higher
Hypertensive crisis	above 180	**and/or**	above 120

LESSON 7: ANTI-INFLAMMATORY DRUGS

Inflammation, the COX pathways, and the NSAIDs: aspirin, paracetamol and the rest

PHARMACOLOGY | FINALS: ANTIHYPERTENSIVE, ANTI-INFLAMMATORY AND GI DRUGS

How do I keep the anti-inflammatory drugs straight?

Inflammation is **prostaglandins at work**: **arachidonic acid → COX → PGE2 and PGI2** (vasodilation, leaky vessels, pain, fever). **Every NSAID is a COX inhibitor**, so the only things to learn are **which COX they block, what they are good for, and what they damage** (stomach, heart, kidney). Dr. Orig's evals: **anti-inflammatory drugs only**.

INFLAMMATION

- The immune system's **protective response** to an injurious stimulus; triggered by **noxious agents, infections, physical injuries**
- **Transient local vasodilation** and **increased capillary permeability → leukocytes and phagocytes** move in and eat the bacteria → **resolution**
- In **diabetics and hypertensives** it may never resolve (diabetic foot)

FIVE CARDINAL SIGNS (MEMORIZE THE LATIN)

- **Calor** – heat | **Rubor** – redness (both from **vasodilation**)
- **Tumor** – swelling (**permeability**: fluid and WBCs leave) | **Dolor** – pain (**nociceptors**) | **Functio laesa** – loss of function

PROSTANOIDS: THE NUMBER ONE PLAYERS

- **Phospholipase A2** frees **arachidonic acid** (a fatty acid) from membrane phospholipids
- **COX → prostaglandins** (and thromboxane) | **LOX → leukotrienes** (allergy, asthma; **montelukast** blocks them)
- **PGE2, PGI2** – primary mediators: **vasodilation, permeability, leukocyte infiltration** | **PGD2** – minor, allergic

PAIN AND FEVER

- **Nociceptors** sense pain (heat, acid, pressure); **prostaglandins make them more sensitive**
- **Hypothalamus** keeps the set point (about 37 °C). **Pyrogens – IL-1, IL-6, TNF-alpha →** hypothalamus makes **PGE2 →** set point rises → **fever** (faster blood flow and WBC recruitment)
- The elderly and immunocompromised may not fever

> **Note:** Dr. Orig: a **tolerable fever** is your body working; treat it only if it is **deleterious** (about 40 °C, rigors) or you are uncomfortable.

NSAIDS

- **Non-steroidal** anti-inflammatory drugs (steroids are anti-inflammatory too: another lecture)
- **Non-selective** (COX-1 and COX-2) or **COX-2 selective**; **all are COX inhibitors**, so prostaglandins fall
- **COX-1** – **constitutive**, housekeeping: **GI mucosal protection, renal blood flow, platelets**
- **COX-2** – **inducible**, the source of prostanoids for **inflammation and cancer**

USES

- **Inflammation** – RA, osteoarthritis, gout (symptomatic) | **Pain** – **low to moderate** inflammatory pain; **primary dysmenorrhea** (first line; start before the flow); migraine | **Fever** – only if deleterious
- **Severe pain** (stage 4 cancer) needs **opioids**, not NSAIDs; opioids can cause addiction, so use NSAIDs first for low-to-moderate pain

ADVERSE EFFECTS

- **GI** – the **most common** (nausea, vomiting, pain); **ulcers** that bleed, perforate or obstruct, because prostaglandin blood flow to the lining is lost
  - **Never give an NSAID in GI bleeding**; **ask first** (blood in stool, vomit or nose), even before aspirin in an MI
- **Cardiovascular** – **coxibs** leave **TXA2** (prothrombotic, atherogenic) unopposed → **MI, stroke**

- **Blood pressure** – about **5%** develop hypertension | **Renal** – **analgesic nephropathy**, slowly progressive renal failure

THE NSAIDS ONE BY ONE

SALICYLATES

- **Aspirin (acetylsalicylic acid)** – **irreversible** COX inhibitor; analgesic-antipyretic **325–1,000 mg every 4–6 h**; loaded in **MI**
- **Salicylates**: aspirin, salicylic acid, methyl salicylate, **sulfasalazine** (RA), **mesalamine** (IBD)
- **Salicylic acid** – an **exfoliant**: wart and corn plasters, skin care

ACETAMINOPHEN (PARACETAMOL)

- Raises the pain threshold; **antipyretic**; **only weak anti-inflammatory**; non-selective COX inhibitor; no prescription needed
- **325–650 mg every 4–6 h**; **max 4 g a day** (2 g in chronic alcoholics) = **8 tablets of 500 mg**
- **NAPQI** → **hepatocyte necrosis**; **10–15 g** in one dose is hepatotoxic, **20–25 g** potentially fatal
- **Hidden in cold remedies**: **Neozep** (+ phenylephrine), **Bioflu** (+ phenylephrine + chlorphenamine), **Tuseran** (+ dextromethorphan + phenylpropanolamine), **Rexidol and Saridon** (+ caffeine), **Skelan** (+ naproxen). **Don't stack them.**

THE OTHER NSAIDS

- **Diclofenac** – potent; long-term arthritis, dysmenorrhea, migraine; also a **gel** for painful joints
- **Ketorolac** – potent analgesic, only moderately anti-inflammatory; **oral, IV, IM**; a favorite of **surgeons**
- **Propionic acids** – **ibuprofen** (Advil, Medicol; **200 mg or less** without prescription; **400 mg every 4–6 h**), **naproxen** (Flanax, Skelan), **ketoprofen**
- **Fenamates** – **mefenamic acid** (Ponstan, Dolfenal); dentists and OB-GYN like it; **no advantage** over other NSAIDs

COX-2 SELECTIVE INHIBITORS (COXIBS)

- **Celecoxib, etoricoxib** – chosen for patients with **GI problems**; **not stronger** painkillers
- **Avoid** in patients prone to **cardiovascular or cerebrovascular disease** (history of **MI or stroke**)

> **Note:** Besides **aspirin** (strong) and **paracetamol** (weak anti-inflammatory), the NSAIDs are **alike**: one is not inherently better.

SUMMARY TABLE: FIVE CARDINAL SIGNS

Latin	English	Cause
Calor	heat	**vasodilation**: more blood to the site
Rubor	redness	**vasodilation**
Tumor	swelling	**increased capillary permeability**: fluid and WBCs leave the vessel
Dolor	pain	**nociceptors** activated and sensitized by prostaglandins
Functio laesa	loss of function	heat, redness, swelling and pain together

SUMMARY TABLE: COX-1 AND COX-2

Feature	COX-1	COX-2
Expression	**constitutive** (always on)	**inducible** (by inflammation)
Role	**housekeeping**: GI mucosal protection, renal blood flow, platelet aggregation	prostanoids for **inflammation and cancer**
Products	TXA2 and protective prostaglandins	**PGE2** and **PGI2**
Blocked by	**non-selective** NSAIDs only	non-selective NSAIDs **and coxibs**

SUMMARY TABLE: NSAIDS AT A GLANCE

Group	Drugs (brands)	Key points
Salicylates	**aspirin**, salicylic acid, methyl salicylate, sulfasalazine, mesalamine	aspirin is **irreversible**; 325–1,000 mg every 4–6 h; mesalamine IBD, sulfasalazine RA, salicylic acid skin
Acetaminophen	**paracetamol** (Biogesic, Neozep, Bioflu, Tuseran, Rexidol, Saridon)	pain and fever; **weak** anti-inflammatory; max **4 g/day**; **NAPQI** hepatotoxicity
Propionic acids	**ibuprofen** (Advil, Medicol), **naproxen** (Flanax, Skelan), ketoprofen	non-selective; ibuprofen 400 mg every 4–6 h, OTC at 200 mg or less
Fenamates	**mefenamic acid** (Ponstan, Dolfenal)	no advantage over other NSAIDs; dysmenorrhea, dental pain
Others	**diclofenac** (also gel), **ketorolac** (oral, IV, IM)	diclofenac potent; ketorolac a potent analgesic, moderate anti-inflammatory
Coxibs	**celecoxib**, **etoricoxib**	for **GI problems**; avoid after **MI or stroke**; not stronger painkillers

SUMMARY TABLE: NSAID ADVERSE EFFECTS

System	Effect	Why
GI	nausea, vomiting, pain; ulcer, bleeding, perforation, obstruction	COX-1 prostaglandins no longer protect the lining; never give in GI bleeding
Cardiovascular	MI, stroke (coxibs)	TXA2 from COX-1 is **prothrombotic** and not blocked
Blood pressure	hypertension in about **5%**	less prostaglandin vasodilation and renal blood flow
Renal	**analgesic nephropathy**	slowly progressive renal failure from chronic use

SUMMARY TABLE: FEVER, PAIN AND THE PROSTANOIDS

Item	Key point
Pyrogens	**IL-1, IL-6, TNF-alpha** → hypothalamus → **PGE2** → higher set point
Nociceptors	nerve fibers that sense pain; **prostaglandins make them more sensitive**
PGE2, PGI2	primary inflammatory prostanoids: vasodilation, permeability, leukocyte infiltration
PGD2	minor, allergic
Leukotrienes	from **LOX**, allergy and asthma; **montelukast** blocks them

LESSON 8: GI DRUGS (DRAFT)

Acid and ulcer drugs, then the laxatives

PHARMACOLOGY | FINALS: ANTIHYPERTENSIVE, ANTI-INFLAMMATORY AND GI DRUGS

How do I keep the GI drugs straight?

Two families. **Acid-peptic drugs** lower acid or protect the lining: **antacids, H2 blockers (-tidine), PPIs (-prazole), sucralfate, misoprostol, bismuth**. **Laxatives** treat constipation: **bulk-forming, stool softeners, osmotic, stimulant**. Learn the **name endings**, the **mnemonics** (CRFN, DROPLE, PMB, PiMP, DGM, LP) and the one **adverse effect** each group is known for.

Draft: made before the lecture

This lesson was made from Dr. Orig's GI Drugs deck before the lecture. **It will be revised after the discussion.**

THE GI TRACT AND THE STOMACH

- **Tract**: mouth, pharynx, esophagus, **stomach**, small intestine (**duodenum, jejunum, ileum**), large intestine, anus
- **Six processes** (**ISMDAD**): **I**ngestion, **S**ecretion, **M**otility, **D**igestion, **A**bsorption, **D**efecation
- **Accessory organs**: teeth, tongue, salivary glands, **liver, gallbladder, pancreas**
- **Stomach**: J-shaped, the **most distensible** part; joins the esophagus to the **duodenum**; mixing chamber, food reservoir, secretes gastric juice and gastrin
- **4 regions**: **cardia** (around the esophageal opening), **fundus** (rounded, above and left of the cardia), **body**, **pylorus** (pyloric antrum, pyloric canal, pylorus)
- **Curvatures**: **lesser** = concave medial; **greater** = convex lateral
- **Lower esophageal sphincter** (gastroesophageal junction) keeps **acid out of the esophagus**; the **pyloric sphincter** guards the way to the duodenum

STOMACH CELLS

- **Surface mucous and mucous neck cells** – **mucus** lining that protects from acid
- **Parietal cell** – **HCl** and **intrinsic factor** (needed for **vitamin B12** absorption)
- **Chief cell** – **pepsinogen** (made **pepsin** by stomach acid) and **gastric lipase**
- **G cell** – **gastrin**, which induces acid secretion | **D cell** – **somatostatin**, which inhibits acid | **ECL cell** – releases **histamine**
- **Parietal cells** sit in the **fundus**; **G cells** in the **antrum**

ACID SECRETION

- The parietal cell has receptors for **gastrin (CCK-B)**, **histamine (H2)** and **acetylcholine (M3**, vagus, rest-and-digest). All three drive the **proton pump (H+/K+-ATPase)**
- **M3 and CCK-B** work through **Gq (IP3/Ca2+)**; **H2** through **Gs (cAMP)**
- **Inhibitors**: **somatostatin** (D cells) and **prostaglandins**
- **Histamine** acts **only directly** on parietal cells (H2). **ACh and gastrin** act **directly and indirectly**, by making **ECL cells** release histamine
- On the figure: **atropine blocks M3**, **cimetidine blocks H2**, **omeprazole blocks the proton pump**

GERD AND PEPTIC ULCER DISEASE

- **GERD** – the **LES opens** and stomach contents and acid back up into the esophagus (**inflamed esophagus**). Allergy is **H1**, not H2
- **PUD** – **gastric and duodenal ulcers**; they **bleed** (vomiting blood, bloody stool)
- **Damaging factors** – gastric acidity, peptic enzymes, **H. pylori**, **NSAIDs**, tobacco, alcohol, duodenal-gastric reflux
- **Protective factors** – surface **mucus**, **bicarbonate** in the mucus, mucosal **blood flow**, epithelial barrier and regeneration, **prostaglandins**. **Ischemia, shock and NSAIDs** weaken them

- **H. pylori** infects the **antrum** (lower stomach), causes **gastritis** (often silent) and then an **ulcer**; from contaminated or uncooked food

DRUGS FOR ACID-PEPTIC DISEASE

- Six classes: **antacids**, **H2 receptor antagonists**, **proton pump inhibitors**, **sucralfate**, **prostaglandin analogs**, **bismuth compounds**

ANTACIDS

- Non-prescription relief of **intermittent heartburn and dyspepsia**; **chemical antagonism** (they neutralize acid). One **156 mEq** dose 1 h after a meal works for up to **2 h**
- **Sodium bicarbonate** (baking soda) – **CO2** forms → **bloating and belching**; can cause **metabolic alkalosis**
- **Calcium carbonate** – **reacts more slowly** than NaHCO3; may cause belching. With **dairy**, Na or Ca bicarbonate can cause **milk-alkali syndrome** (**hypercalcemia**, **renal insufficiency**): don't take antacids with milk. Brand: **Gaviscon Double Action**
- **Magnesium hydroxide / aluminum hydroxide** – no gas, no belching. **Magnesium = diarrhea** (Mg laxatives are used before endoscopy); **aluminum = constipation**. Given together they **cancel each other**. Brand: **Kremil-S** (aluminum hydroxide, magnesium hydroxide, simeticone)

H2 RECEPTOR ANTAGONISTS (-TIDINE)

- **Cimetidine, ranitidine, famotidine, nizatidine** (**CRFN**); **competitive** block of the parietal **H2 receptor**; they cut **basal and meal-stimulated** acid and do **not** touch H1 or H3. **Nizatidine** has the least first-pass metabolism
- **Potency**: **famotidine** is the most potent (smallest dose); **cimetidine** the least
- **Uses**: **GERD** (infrequent heartburn or dyspepsia, under 3 times a week), **PUD** (largely replaced by PPIs), nonulcer dyspepsia, preventing bleeding from **stress-related gastritis**
- **Extremely safe. Cimetidine**: mental status changes, **gynecomastia and impotence in men** (raises estrogen). Brand: **Ratidin** (ranitidine)

PROTON PUMP INHIBITORS (-PRAZOLE)

- **Omeprazole, esomeprazole, lansoprazole, dexlansoprazole, rabeprazole, pantoprazole** (**DROPLE**)
- **Inactive prodrugs**, taken about **1 h before a meal** (usually breakfast), **once a day**. Acid stays down for up to **24 h** because the pump is **irreversibly inactivated**: the body must make new H+/K+-ATPase
- **Uses**: **GERD** (most effective, intermittent courses), **PUD** (faster relief and ulcer healing than H2RAs), nonulcer dyspepsia, **stress-related mucosal bleeding**, **gastrinoma**
- **Adverse effects**: **vitamin B12 deficiency**; more **respiratory and enteric infections** (stomach acid protects against microbes). PPIs are now the ones mostly given

H. PYLORI AND GASTRINOMA

- **H. pylori** is a common cause of ulcer: **antibiotics kill it, the PPI heals the ulcer**. First-line regimens all include a **PPI**
- **Clarithromycin triple** – clarithromycin + amoxicillin or metronidazole | **Bismuth quadruple** – bismuth + metronidazole + tetracycline | **Nonbismuth quadruple** – amoxicillin + clarithromycin + metronidazole (the one given in the PH since available)
- **Gastrinoma (Zollinger-Ellison syndrome)** – a pancreatic or duodenal tumor secretes **gastrin** → too much acid → multiple stubborn ulcers in unusual places like the **jejunum**. Treatment: **surgery** to remove the tumor plus a **PPI**

SUCRALFATE

- Salt of **sucrose** complexed to **sulfated aluminum hydroxide**; forms a **viscous, tenacious paste** that binds selectively to **ulcers and erosions** for up to **6 h**; taken on an **empty stomach** (1 h before meals)

MISOPROSTOL (PROSTAGLANDIN ANALOG)

- An analog of **PGE2**; **lowers gastric acid**; taken 3–4 times a day; **reduces NSAID-induced ulcers**
- Can cause **abortion**: **not for pregnant patients**

BISMUTH COMPOUNDS

- **Coat** ulcers and erosions; stimulate **prostaglandin, mucus and bicarbonate** secretion (**PMB**); **direct antimicrobial** activity against **H. pylori**. Brand: **Pylera** (bismuth subcitrate potassium, metronidazole, tetracycline)

Note: Memory tip: CRFN = the -tidines, DROPLE = the -prazoles, PMB = what bismuth makes the stomach release. **Kremil-S cancels itself out**: Mg loosens, Al hardens.

LAXATIVES

- Given to patients with **constipation**. Four groups: **bulk-forming, stool softeners, osmotic, stimulant**

BULK-FORMING (PIMP)

- **Fiber supplements**: **psyllium** (brand **Metamucil**), **methylcellulose, polycarbophil**; they **absorb water** and form a bulky, emollient **gel** that distends the colon and **promotes peristalsis**

STOOL SOFTENERS (DGM)

- **Docusate** (oral or enema), **glycerin suppository**, **mineral oil**; soften the stool so **water and lipids penetrate**; oral or rectal
- **Docusate** is a stimulant-free softener. Brand: **Dulcolax Stool Softener**

OSMOTIC (LP)

- **Lactulose, polyethylene glycol (PEG)**: soluble but **nonabsorbable**, so more fecal fluid and more liquid stool
- **PEG** gives **complete colonic cleansing** before GI endoscopy or colonoscopy (brand **SureLax**, PEG 3350). PEG plus **bisacodyl (Dulcolax)** before endoscopy

STIMULANT

- They **directly stimulate or irritate the GI lining**
- **Anthraquinone derivatives** – aloe, **senna**, cascara (brand **Senokot Forte**, calcium sennosides); chronic senna use causes **melanosis coli** (a dark, pigmented colon). Brand: **Biofitea** (senna)
- **Diphenylmethane derivatives** – **bisacodyl**; works in **6–10 h** by mouth, **30–60 min** as a suppository. Brand: **Dulcolax suppository**

Note: Watch the trap: Dulcolax is the brand of both the **docusate softener** and the **bisacodyl stimulant**.

Name endings at a glance

-tidine H2 blocker · **-prazole** PPI · **psyllium, methylcellulose, polycarbophil** bulk-forming · **docusate, glycerin, mineral oil** softeners · **lactulose, PEG** osmotic · **senna, cascara, bisacodyl** stimulant

SUMMARY TABLE: DRUG CLASSES FOR ACID-PEPTIC DISEASE

Class	Drugs	How it works	Key points
Antacids	**sodium bicarbonate**, **calcium carbonate**, **magnesium hydroxide / aluminum hydroxide**	chemical antagonism: neutralize acid	non-prescription; intermittent heartburn and dyspepsia
H2 receptor antagonists (-tidine)	**cimetidine**, **ranitidine**, **famotidine**, **nizatidine**	competitive block of the parietal **H2 receptor**	**famotidine** most potent; **cimetidine** gynecomastia
Proton pump inhibitors (-prazole)	**omeprazole**, **esomeprazole**, **lansoprazole**, **dexlansoprazole**, **rabeprazole**, **pantoprazole**	**irreversibly** inactivate the **H+/K+-ATPase**	inactive prodrugs; **once a day**, 1 h before breakfast
Sucralfate	**sucralfate**	viscous paste that **binds ulcers and erosions**	up to **6 h**; empty stomach
Prostaglandin analog	**misoprostol** (PGE2 analog)	lowers gastric acid	**NSAID ulcers**; **not in pregnancy**
Bismuth compounds	bismuth (**Pylera**)	coat ulcers; ↑ **prostaglandin, mucus, bicarbonate**	**kill H. pylori** directly

SUMMARY TABLE: ANTACIDS COMPARED

Antacid	Chemistry	Effects to remember	Brand on the slide
Sodium bicarbonate	baking soda; forms **CO2**	**gastric distention and belching**; can cause metabolic alkalosis	–
Calcium carbonate	H+ binds CO3	**reacts more slowly** than NaHCO3; belching; dairy → **milk-alkali syndrome** (hypercalcemia, renal insufficiency)	**Gaviscon Double Action**
Magnesium hydroxide	H+ binds OH	no gas; **diarrhea** (laxative)	**Kremil-S** (with aluminum hydroxide and simeticone)
Aluminum hydroxide	H+ binds OH	no gas; **constipation**	**Kremil-S**

SUMMARY TABLE: H2 BLOCKERS VERSUS PPIS

Feature	H2 receptor antagonists	Proton pump inhibitors
Target	**H2 receptor** on the parietal cell (competitive)	**H+/K+-ATPase** (irreversible)
Drugs	**cimetidine, ranitidine, famotidine, nizatidine** (CRFN)	**omeprazole, esomeprazole, lansoprazole, dexlansoprazole, rabeprazole, pantoprazole** (DROPLE)
Uses	GERD (infrequent), PUD (largely replaced), nonulcer dyspepsia, stress gastritis bleeding	GERD (**most effective**), PUD (**faster healing**), nonulcer dyspepsia, stress mucosal bleeding, **gastrinoma**
Adverse effects	extremely safe; **cimetidine**: mental status changes, gynecomastia, impotence	**vitamin B12 deficiency**; more respiratory and enteric infections

SUMMARY TABLE: H. PYLORI REGIMENS (BY DRUG)

Regimen	Drugs (each with a PPI)
Clarithromycin triple	**clarithromycin** + **amoxicillin** or **metronidazole**
Bismuth quadruple	**bismuth** + **metronidazole** + **tetracycline**
Nonbismuth quadruple	**amoxicillin** + **clarithromycin** + **metronidazole** (given in the PH since available)

SUMMARY TABLE: LAXATIVES

Group	Drugs	How it works	Key points
Bulk-forming (PiMP)	**psyllium**, **methylcellulose**, **polycarbophil**	absorb water; bulky gel distends the colon, **promotes peristalsis**	–
Stool softeners (DGM)	**docusate**, **glycerin suppository**, **mineral oil**	soften stool: water and lipids penetrate	oral or rectal; **Dulcolax Stool Softener** = docusate
Osmotic (LP)	**lactulose**, **polyethylene glycol (PEG)**	soluble, nonabsorbable: more **fecal fluid**	PEG: **colon cleansing** before endoscopy
Stimulant	**aloe, senna, cascara** (anthraquinones); **bisacodyl** (diphenylmethane)	induce a bowel movement	senna: **melanosis coli**; bisacodyl **6–10 h** orally, **30–60 min** rectally

END OF LESSON